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Organizational Readiness, AI Literacy, and the New Frontier of R&D: How Generative AI Shapes Innovation Capacity

Stefanie Steinhauser  | Paulina Heid

Technical University of Applied Sciences Amberg-Weiden, Weiden, Germany

Correspondence: Stefanie Steinhauser (s.steinhauser@oth-aw.de)

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ABSTRACT

Generative artificial intelligence (AI) can transform research and development (R&D) by enhancing efficiency, accelerating product development, and fostering innovation. As companies seek to integrate this technology, understanding the role of organizational readiness and AI literacy as well as generative AI's impact on innovation capacity becomes increasingly important. This study examines whether cultural and structural readiness and AI literacy affect the use of generative AI and how generative AI influences R&D activities and innovation capacity. The study employs a qualitative pre-study and survey data, analyzed with structural equation modeling. The findings show that cultural readiness, reflecting an organization's openness to AI-driven change, is a key enabler of successful adoption. Structural readiness, defined by the technological and procedural alignment with AI applications, appears to play a conditional role, depending on the level of employees' AI literacy. The findings also demonstrate that generative AI supports R&D processes and, in turn, enhances innovation capacity. By providing insights into the interplay between organizational readiness, AI literacy, generative AI utilization, and its impact on R&D activities and innovation capacity, this research advances the academic understanding of how generative AI can contribute to innovation processes and provides practical implications for organizations aiming to leverage generative AI for competitive advantage.

1 | Introduction

"[T]here is a long way before AI platformssuch as ChatGPT could be capable to lead independently to meaningful product, process, or business model innovation. At best they can help enhance human intelligence for innovation by augmenting human intelligence. As AI platforms and the underlying technology will evolve, future research will need to investigate if and to what extent the role played by generative AI will be increasingly relevant in triggering innovation outcomes."

(Dwivedi et al. 2023, 6).

While this statement points to the unknown future of generative artificial intelligence (AI) in companies, technological innovations are one of the key factors driving economic growth. Selected innovations of fundamental importance are described by Brynjolfsson and McAfee (2017) as so-called general-purpose technologies. These include the steam engine, electricity, and the internal combustion engine, as each triggered a stimulus for subsequent innovations and opportunities that complement each other. AI is now also mentioned in the same context: AI may be the most important general-purpose technology in the modern era (Brynjolfsson and McAfee 2017). In addition to putting AI on par with previous revolutionary technologies, AI is expected to have an immense impact on the global economy.

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Companies that expand their technological capabilities and promote their strategic investments in innovation can expect a 60% revenue growth rate and 40% profitability. However, only 3% of leading companies currently have this capability. Companies with industry-leading digital core elements will reinvent twice as many functionalities with generative AI compared to other companies and are expected to generate twice the added value (Narain et al. 2024). Generative AI is an indispensable tool for research and development (R&D) in companies. Early adopters of AI, such as GE, Siemens, Pfizer, and Nestlé, report substantial benefits, including significant reductions in development timelines and an accelerated rate of innovation (Cooper and Brem 2024). Major companies such as BASF or Unilever are already using generative AI to identify the optimal proportions of molecules or to develop enzymes (Müller 2022; Unilever 2023). In the field of design, it is assumed that up to 30%–50% of the development time and up to 20% of the costs can be saved by using generative AI algorithms (Brossard et al. 2020). The utilization of AI also lowers search costs and increases the speed of R&D processes (Johnson et al. 2022).

Companies see generative AI as a tool to improve productivity, efficiency, and decision-making, reduce time-to-market and mistakes, and increase agility, with the long-term winners being those companies that can differentiate themselves from the competition by using this technology, enabling a comprehensive change in the company and thus creating value in a new way (Mittal et al. 2024; Cooper and Brem 2024). However, although AI has attracted significant attention and early adopters have demonstrated its potential, its use in R&D remains surprisingly limited in European and US firms (Cooper and Brem 2024). Understanding how organizational readiness and AI literacy shape the effective utilization of AI is therefore critical for enabling firms to realize its potential impact on R&D and innovation capacity. Hence, this study aims to gain insights into these relationships. We employ a qualitative pre-study and a quantitative survey, which we analyze with structural equation modeling to explore these relationships.

In doing so, our research contributes to the academic discourse on organizational readiness, AI literacy, generative AI, R&D activities, and innovation capacity by integrating perspectives from change management, information systems (IS), utilization of generative AI, and innovation research. First, it advances the understanding of organizational readiness by examining its role in the utilization of generative AI, distinguishing between cultural and structural dimensions. While previous research has largely focused on readiness for traditional IT systems (e.g., Shahrabi and Paré 2015), this study extends the concept to the emerging field of generative AI, providing a foundation for future research on AI-driven transformation. Second, by integrating AI literacy, which has primarily been studied in IS research, into R&D and innovation management research, this paper provides a new perspective as well as new insights into the interaction of organization- and individual-level factors in this context. Moreover, it adds to the presently limited understanding of how AI literacy translates into specific actions when users engage with AI (Colville and Ostern 2024), showing that varying AI literacy levels shape how employees collaborate with generative AI and incorporate its output into R&D work. Third, it enriches the literature on generative AI in

R&D by exploring how generative AI can reshape development processes, idea generation, and product innovation. By situating generative AI within the broader context of R&D activities, this study offers new perspectives on the intersection of generative AI and innovation. Fourth, this research contributes to studies on innovation capacity by investigating how AI-driven advancements in R&D influence an organization's ability to generate and implement novel ideas.

2 | Theoretical Background

2.1 | Generative Artificial Intelligence

AI represents the higher-level instance of machine learning (ML) that performs tasks requiring human intelligence (Kulkarni et al. 2023). Within ML, deep learning (DL) is central, as both approaches can identify hidden patterns in data and apply models to new situations (e.g., Murphy 2012). DL relies on artificial neural networks, which, similar to the human brain, consist of nodes (neurons) and edges (synapses). Their multi-layered structure enables the recognition of complex and abstract relationships. Generative AI is a specialized branch of DL: it uses large datasets for training and analysis, but its key distinction lies in generating new and unique content (e.g., Feuerriegel et al. 2024).

Traditionally, generative AI was seen as suitable mainly for analytical tasks (Kakatkhar et al. 2020), carried out by specialists (Bilgram and Laarmann 2023), while creative applications were not associated with the technology (Füller et al. 2022). This changed with the release of ChatGPT, a large language model (LLM) within the “generative pretrained transformer (GPT)” family (Bilgram and Laarmann 2023), introduced by OpenAI in November 2022 (OpenAI 2022). Generative AI such as ChatGPT can analyze and generate text, understand conversational context, and provide tailored responses, enabling human-like interaction through advanced natural language processing (NLP). Its intuitive usability allows even inexperienced users to apply it effectively, for example, by converting text descriptions into code (Bilgram and Laarmann 2023; Deng and Lin 2022). Other models extend generative AI beyond text: generative adversarial networks (GANs) generate images by training a generator and discriminator until the generator produces realistic results (Aggarwal et al. 2021). Today, providers also enable the generation of videos, 3D worlds, audio, or websites via text prompts (De Ridder et al. 2023). These applications are accessible to both private and commercial users. This research, however, focuses on the commercial use of generative AI.

2.2 | Organizational Readiness

The implementation of innovations in companies is a challenging and current issue. In addition to political, socio-cultural, and economic motives that trigger internal change, advancing digitalization is also considered a driving force that companies should proactively respond to in order to remain competitive (e.g., Deszca et al. 2020). The concept of organizational readiness for change originates from the change management literature (e.g., Coch and French 1948; Lewin and Cartwright 1964; Lewin 1947).

Organizational readiness is considered a necessary factor that increases the likelihood of successful organizational change (Rafferty et al. 2013; Gräfe and Kauffeld 2024): “*assessing readiness and then subsequently increasing efforts to create [...] organizational change readiness may be the necessary ingredient to increase the likelihood of successful organizational change*” (Rafferty et al. 2013, 130). Since its introduction, organizational readiness has been examined across numerous disciplines (Shahrasbi and Paré 2014), such as sales (Weeks et al. 2004), healthcare management (Weiner et al. 2008), information systems (Iacovou et al. 1995), and human resources (Eby et al. 2000).

The term organizational readiness is defined in diverse ways. On the one hand, extant literature refers to the individual or organizational level. On the other hand, approaches differ on whether readiness is psychological or structural (Weiner et al. 2020). Based on literature reviews by Shahrasbi and Paré (2014) and Weiner et al. (2020), Table 1 provides a selection of key definitions.

Shahrasbi and Paré (2014) identified six categories of organizational readiness for IS research: financial, technological, personnel, process and operational, cultural, and business readiness. In this study, we focus on structural readiness and cultural readiness to analyze the integration of generative AI in R&D. Structural readiness encompasses the provision of technical resources (Iacovou et al. 1995) and compatible work practices (Sutanonpaiboon and Pearson 2006), while cultural readiness refers to shared values and a positive culture toward IS implementation (Shahrasbi and Paré 2014). We focus on these two because they most directly shape employees' ability and willingness to use generative AI in R&D, whereas other dimensions operate at a broader strategic level. Moreover, structural and cultural readiness represent the two key facets of organizational readiness—structural and psychological—offering a comprehensive, multidimensional approach particularly suited for IT-based change (Shahrasbi and Paré 2014; Shahrasbi and Paré 2015).

Organizational readiness can be understood as a core micro-foundation of absorptive capacity and, by extension, dynamic capabilities (Cohen and Levinthal 1990; Teece et al. 1997; Felin et al. 2015) (see also Table 2). Absorptive capacity, viewed as a dynamic capability, depends on baseline resources and

conditions that enable organizations to acquire, assimilate, and apply new knowledge (Zahra and George 2002). Among the various dimensions of readiness, structural readiness provides the necessary infrastructure and formal processes that lower barriers to the effective use of new technologies, while cultural readiness fosters openness, shared values, and support for experimentation. Together, these dimensions create the organizational context in which dynamic capabilities of sensing, seizing, and reconfiguring can be effectively deployed (e.g., Montasser et al. 2023; Adhiatma et al. 2022). In the absence of such readiness, organizations struggle to mobilize employee skills, technologies, and routines, limiting their ability to translate generative AI adoption into adaptive innovation outcomes.

2.3 | AI Literacy

So far, research on AI literacy has primarily emerged from IS literature. AI literacy refers to an individual's comprehensive proficiency in relation to AI, encompassing the ability to critically use and evaluate AI systems, as well as to communicate and collaborate with AI effectively (Long and Magerko 2020). AI literacy presents a critical factor in shaping how users such as employees engage with generative AI (Colville and Ostern 2024). Despite its growing importance, there is a focus on students or the general public rather than on employees and business contexts in education as well as in research (e.g., Cetindamar et al. 2024). While prior studies have largely centered on users (Wang et al. 2023), students (Steinbauer et al. 2021), developers (Sambasivan et al. 2021), and executives (Pinski et al. 2024), the perspectives of employees in R&D remain underexplored, leaving a critical gap in understanding AI's role in organizational innovation.

AI literacy presents a crucial individual-level factor that can influence the utilization of generative AI and its impact on R&D processes and innovation capacity. Technology acceptance literature (see Table 2) studies factors that affect use of new technologies by individuals. Experience of an individual in a broader sense presents a related construct (Pinski et al. 2024) and is integrated into UTAUT as a moderator of facilitating conditions (i.e., “the degree to which an individual

TABLE 1 | Comparison of the definitions of organizational readiness (based on Weiner et al. 2020).

Definition	Content	Level
<i>“As a working definition, we consider organizational readiness for change as the extent to which organizational members are psychologically and behaviorally prepared to implement organizational change.”</i> (Weiner et al. 2008)	- Psychological approach - Employees considered as a whole/unit	Organization
<i>“[T]he extent to which employees hold positive views about the need for organizational change (i.e., change acceptance), as well as the extent to which employees believe that such changes are likely to have positive implications for themselves and the wider organization.”</i> (Jones et al. 2005)	- Emotion - Persuasion of the individual - Positive development caused by change	Organization Individual
<i>“[It] collectively reflects the extent to which an individual or individuals are cognitively and emotionally inclined to accept, embrace, and adopt a particular plan.”</i> (Holt et al. 2007)	- Emotion - Based on individual understanding of the change	Individual

TABLE 2 | Related theories.

Theory	Description	Level
Absorptive capacity	<i>“the ability of a firm to recognize the value of new, external information, assimilate it, and apply it to commercial ends is critical to its innovative capabilities. We label this capability a firm’s absorptive capacity [...]”</i> (Cohen and Levinthal 1990, 128) <i>“Our definition suggests that the four organizational capabilities of knowledge acquisition, assimilation, transformation, and exploitation build on each other to yield ACAP [absorptive capacity]—a dynamic capability that influences the firm’s ability to create and deploy the knowledge necessary to build other organizational capabilities (e.g., marketing, distribution, and production). These diverse capabilities give the firm a foundation on which to achieve a competitive advantage that yields superior performance [...]”</i> (Zahra and George 2002, 188)	Organization Individual
Dynamic capabilities	<i>“We define dynamic capabilities as the firm’s ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments. Dynamic capabilities thus reflect an organization’s ability to achieve new and innovative forms of competitive advantage [...]”</i> (Teece et al. 1997, 516) <i>“[...] dynamic capabilities can be disaggregated into the capacity (1) to sense and shape opportunities and threats, (2) to seize opportunities, and (3) to maintain competitiveness through enhancing, combining, protecting, and, when necessary, reconfiguring the business enterprise’s intangible and tangible assets.”</i> (Teece 2007, 1319)	Organization Individual
Technology acceptance	Technology acceptance literature investigates the determinants that facilitate or impede individuals’ adoption and use of novel technologies. The Technology Acceptance Model (TAM) encompasses the constructs <i>Perceived usefulness</i> , <i>Perceived ease of use</i> , <i>Attitude toward using</i> , <i>Behavioral intention to use</i> , and <i>Actual system use</i> , and suggests that individuals are more inclined to use a technology when they perceive it as both useful and easy to use (Davis et al. 1989). The Unified Theory of Acceptance and Use of Technology (UTAUT) extends this perspective by incorporating additional determinants, including <i>social influence</i> and <i>facilitating conditions</i> (Venkatesh et al. 2003).	Individual Organization

believes that an organizational and technical infrastructure exists to support use of the system”) (Venkatesh et al. 2003). Yet, research on the impact of experience with a technology on the effect of facilitating conditions is surprisingly sparse and findings are ambivalent (Williams et al. 2015). Moreover, employees’ AI literacy presents an individual-level microfoundation as the absorptive capacity of an organization depends on the absorptive capacities of its individual employees and an organization’s ability to recognize opportunities (i.e., dynamic capabilities) depends on the employees’ capabilities and extant knowledge (Cohen and Levinthal 1990; Teece 2007; Eisenhardt et al. 2010; Cetindamar et al. 2024). Hence, studying the moderating role of AI literacy can provide valuable insights (e.g., Colville and Ostern 2024).

2.4 | Impact of AI on R&D- and Innovation Management-Related Activities

R&D serves as a critical driver of business innovation and technical competitive advantage, enabling companies to create new products¹ and production processes. Through R&D, firms can respond effectively to external factors such as technological advancements, market shifts, and evolving consumer behavior, ensuring sustained competitiveness. Product development, in particular, demands creative and strategic problem-solving, especially when projects begin with unclear objectives or undefined

solutions. Whether enhancing existing products, creating variations, or developing entirely new offerings, organizations rely on diverse methods and creativity techniques to identify optimal solutions. Leveraging the expertise of seasoned employees and researching prior solutions often accelerates problem-solving. Additionally, intuitive approaches, combined with various creativity techniques, are especially effective for generating novel solutions that transcend conventional approaches (e.g., Krishnan and Ulrich 2001).

New technologies such as generative AI offer opportunities to enhance efficiency and effectiveness, acting both as facilitator and originator of innovation (Brem et al. 2023). Figure 1 illustrates the positioning of AI within the new product process.

Generative AI has been found to influence R&D-related human skills and tasks (Giordano et al. 2024). Furthermore, AI can take over essential intermediation activities and search functions and emerge as a non-human innovation intermediary (Just 2024).

Along the R&D process, generative AI supports different stages. In idea generation and early concept development, AI provides creative input through prompts and pre-specified needs, offering written solutions and novel concepts (Bilgram and Laarmann 2023; Garbuio and Lin 2021; Pennefather 2023; PTC 2024). Studies have shown that generative AI can

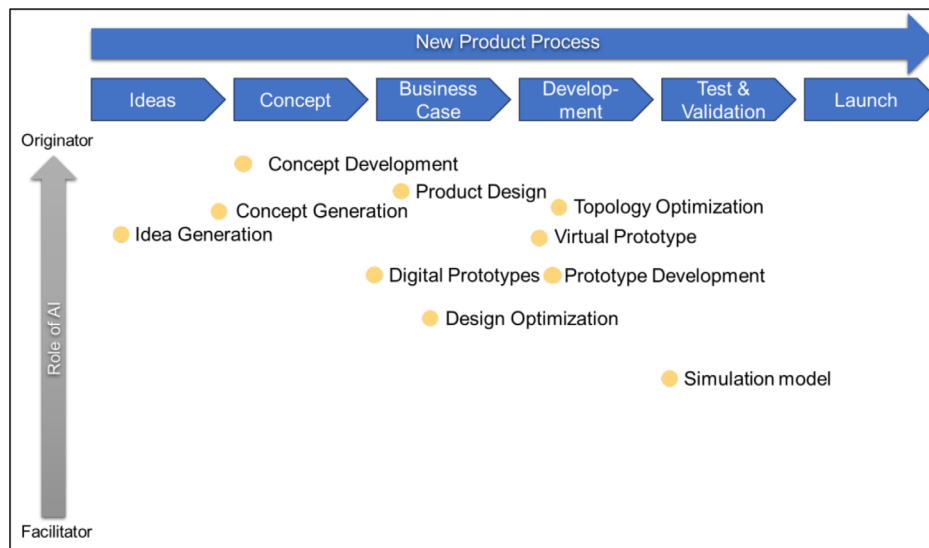


FIGURE 1 | AI positioning in the new product process (based on Cooper and McCausland 2024).

stimulate creativity, expand solution spaces, and generate ideas with high novelty and customer benefit (Joosten et al. 2024; Bouschery et al. 2023). In practice, Bilgram et al. (2023) demonstrated its use in design thinking workshops, where AI accelerated problem-solving and prototyping, while Noy and Zhang (2023) found that AI improved writing quality and speed, especially for less-skilled workers, though verification remained necessary.

In design and prototyping, AI is used for concept design, digital prototyping, and customization, enabling real-time testing and iteration (Cooper 2024; Mariani and Dwivedi 2024). Generative AI can also be combined with CAD tools for topology optimization, improving structure and reducing weight (Walzl and Buchmayr 2017). Studies confirm its potential to inspire designers and push creative boundaries (Pennefather 2023; Garbuio and Lin 2021).

In process optimization and development tasks, AI contributes by reducing simulations, accelerating decision-making, and supporting adaptations. Zimmerling et al. (2019) showed that pre-trained neural networks can cut simulations by 30%–70%, while Füller et al. (2022) highlighted AI's value for trend identification and analytical tasks, though less for idea generation.

Finally, studies on productivity and efficiency show broader impacts of generative AI across knowledge work. Dell'Acqua et al. (2023) found enhanced quality and reduced processing time among strategy consultants when tasks aligned with AI capabilities, though variability increased. Similarly, Noy and Zhang (2023) and Füller et al. (2022) emphasized time savings and performance improvements, but also highlighted limitations related to verification and applicability across contexts.

Together, these studies underscore the transformative role of generative AI across sequential R&D tasks—from idea generation and design to prototyping and process optimization—while also pointing to boundary conditions and the need for further research into its organizational integration.

2.5 | Innovation Capacity

Innovation capacity is a central concept in understanding how organizations can maintain and enhance their competitive advantage. It refers to a company's ability to develop and implement new products, processes, and ideas that create value (e.g., Forsman 2011). Prajogo and Ahmed (2006, 504) define innovation capacity as “the organizational potential to innovate, which is determined by the skills and strengths in basic R&D and technology.” This definition emphasizes that innovation capacity is not simply an abstract concept but a tangible and measurable factor linked to an organization's ability to develop new ideas that lead to valuable products and services. The capacity to innovate depends on having a conducive environment that fosters creativity, supports R&D activities, and encourages collaboration among diverse teams. More efficient and effective product and process development in R&D directly influences innovation capacity, as these activities allow companies to respond to technological changes, market demands, and evolving consumer preferences. Innovations are the engine for growth and employment, which highlights their essential role in the long-term survival and competitiveness of firms.

A defining feature of innovation is its ability to transform novel ideas into marketable outcomes. Innovation capacity is multifaceted, shaped by both organizational factors—such as culture, structure, leadership, and strategy—and technological factors, particularly resources, R&D capabilities, and technological infrastructure (Prajogo and Ahmed 2006). This dual focus underlines its complexity and central role in determining innovative output. It also reflects the ability to adapt to external pressures, such as market shifts or technological advances, and to leverage these for novel solutions. Stronger internal capabilities increase the likelihood of generating value from innovations.

Moreover, innovation capacity is dynamic rather than static. Organizations can strengthen it by investing in new technologies, skills, and processes, aligning R&D with both technological

advancements and a culture that fosters creativity, risk-taking, and continuous improvement. Ultimately, innovation capacity determines a firm's ability to sustain competitiveness by continuously developing products, processes, and services that meet evolving demands. In rapidly changing industries, it is often the key to long-term success.

Extant research has shown that the relationships between innovation stimulus and innovation capacity, as well as between innovation capacity and innovation performance, are significant and strong (Prajogo and Ahmed 2006), that differences in innovation development and innovation capacity between manufacturing and service sectors are only slight (Forsman 2011), that innovation capability strongly drives firm performance across all innovation types (Rajapathirana and Hui 2018), and that the interaction between potential absorptive capacity and innovation capacity positively affects performance (Hurtado-Palomino et al. 2022).

3 | Hypotheses Development

In the following, we develop our hypotheses. Figure 2 presents our research model, which describes the conceptual framework of this study.

3.1 | Organizational Readiness and Utilization of Generative AI

The utilization of generative AI within companies depends significantly on both structural readiness and cultural readiness. Structural readiness ensures the technological foundation and operational alignment needed to integrate generative AI effectively, while cultural readiness fosters a supportive environment that embraces the use of innovative technologies. These factors collectively influence the likelihood of generative AI adoption in an organizational setting. Structural readiness reflects the technological capabilities and the compatibility of existing systems and processes with the requirements of generative AI. A firm with advanced technological infrastructure, such as access to large-scale data, robust IT systems, and AI-compatible workflows, is well-positioned to deploy generative AI tools effectively.

Moreover, such readiness reduces technical barriers, making the adoption of generative AI more feasible. For instance, companies with seamless data integration capabilities and scalable IT frameworks are better equipped to leverage generative AI applications, such as automated content generation, predictive modeling, and design optimization. Therefore, a higher level of structural readiness directly facilitates the integration and usage of generative AI within the organization. Cultural readiness encompasses the shared values, attitudes, and organizational mindset toward AI adoption. A positive culture that emphasizes innovation and embraces technological change can significantly influence the likelihood of generative AI usage. When employees and leaders view AI as an opportunity rather than a threat, they are more likely to advocate for its implementation and actively engage with the technology. Cultural readiness also mitigates resistance to change, ensuring that generative AI is integrated smoothly into daily workflows. For example, organizations that value experimentation and learning are more likely to explore generative AI applications, utilizing its capabilities to foster creativity and efficiency. As a result, cultural readiness serves as a critical enabler, promoting the acceptance and adoption of generative AI across the company. Hence, we postulate:

H1. *The more the structural readiness is geared towards the introduction of AI, the more likely it is that generative AI will be used in the company.*

H2. *The more the cultural readiness is geared towards the introduction of AI, the more likely it is that generative AI will be used in the company.*

3.2 | Utilization of Generative AI and Impact on R&D Activities

The integration of generative AI into R&D activities is anticipated to significantly enhance the efficiency and effectiveness of innovation processes. Generative AI, particularly advanced large language models, expands the scope of problem-solving and ideation within R&D by providing access to a vast reservoir of knowledge. This broader access enables the identification of high-quality solutions and fosters improved innovation performance (e.g., Bouschery et al. 2023). Moreover, generative

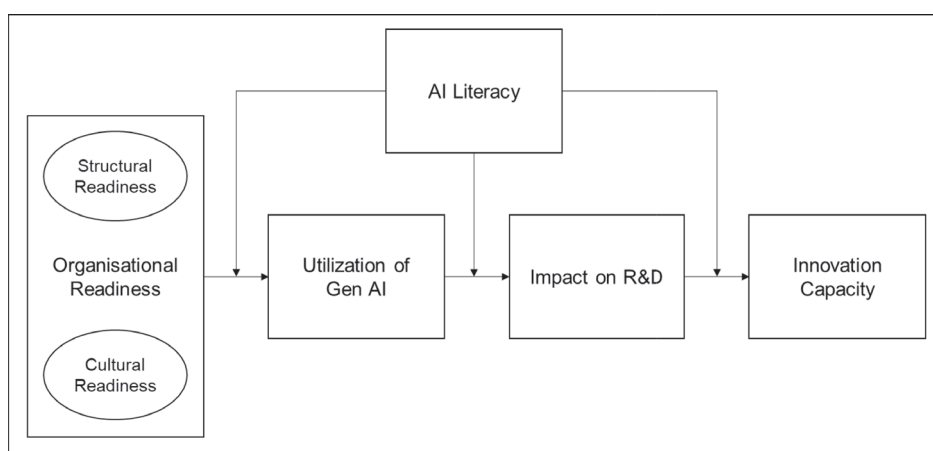


FIGURE 2 | Research model.

AI facilitates real-time experimentation, validation, and prototyping, which accelerates the transformation of ideas into products or processes. By streamlining these traditionally time-intensive tasks, generative AI reduces the time-to-market for innovations and enhances the iterative development process (Mariani and Dwivedi 2024). These capabilities suggest that generative AI can optimize workflows, improve decision-making, and expand the creative potential of R&D teams, thereby positively impacting R&D activities overall. Accordingly, we propose:

H3. *The utilization of generative AI has a positive impact on research and development activities.*

3.3 | Impact on R&D Activities and Innovation Capacity

Generative AI has the potential to significantly enhance R&D activities by improving productivity, efficiency, and the quality of outcomes. These enhancements, in turn, are likely to positively impact the innovation capacity of a company, defined as its ability to develop new products, services, or processes that create value in competitive markets. Generative AI streamlines R&D processes by accelerating product development, facilitating prototyping, or enabling more complex projects. Such improvements increase the overall productivity and efficiency of R&D teams, allowing them to focus on high-value activities that drive innovation (Mittal et al. 2024; Cooper and Brem 2024). Additionally, by providing access to diverse data and generating creative solutions, generative AI expands the problem-solving scope, enabling companies to develop novel and impactful innovations. Moreover, generative AI has been shown to enhance the quality of R&D outcomes. By improving decision-making processes and providing data-driven insights, generative AI helps organizations produce more refined, well-informed, and high-quality results (e.g., Dell'Acqua et al. 2023). This, in turn, enhances the competitiveness and relevance of the innovations generated. These combined effects—enhanced efficiency, productivity, and quality—equip companies with a stronger foundation for sustained innovation. By leveraging generative AI to augment their R&D capabilities, firms are better positioned to create cutting-edge solutions that meet evolving market demands. Therefore, this research hypothesizes that R&D activities enhanced by generative AI have a positive impact on the innovation capability of a company. Hence, we postulate:

H4. *Research and development activities that are enhanced by generative AI have a positive impact on the innovation capacity of a company.*

3.4 | AI Literacy as a Moderator

Aside from organization-level determinants, characteristics of individual employees can also influence the utilization of generative AI and its impact on R&D processes and innovation capacity. Hence, we argue that employees' AI literacy can moderate the investigated effects. Structural readiness reduces barriers to the utilization of generative AI by providing an adequate infrastructure. Employees with low AI literacy are more dependent on such

structures to effectively use AI. By contrast, employees with high AI literacy do not depend on these structures as much and may even perceive rigid structures as constraining, since they can already operate AI tools independently and may prefer flexibility. As a result, AI literacy may weaken the positive effect of structural readiness on utilization. Accordingly, we propose:

H5a. *AI literacy negatively moderates the relationship between structural readiness and utilization of generative AI, such that the positive effect of structural readiness is stronger for employees with low AI literacy and weaker for employees with high AI literacy.*

Cultural readiness fosters employee willingness to use generative AI. For employees with low AI literacy, cultural encouragement can reduce fear and builds confidence, increasing utilization. For employees with high AI literacy, cultural readiness becomes less critical, as they already possess confidence and skills, and may not need cultural reinforcement to adopt AI. Thus, AI literacy may weaken the positive effect of cultural readiness on utilization. We postulate:

H5b. *AI literacy negatively moderates the relationship between cultural readiness and utilization of generative AI, such that the positive effect of cultural readiness is stronger for employees with low AI literacy and weaker for employees with high AI literacy.*

Utilization of generative AI enhances R&D processes by streamlining prototyping, supporting exploration, and broadening problem-solving. Employees with high AI literacy may be better able to unlock these benefits because they can use AI more effectively and integrate it into workflows. In contrast, employees with low AI literacy may underutilize features or misapply tools, limiting the impact of usage on R&D outcomes. As a result, AI literacy may strengthen the positive effect of utilization on R&D processes. Accordingly, we propose:

H5c. *AI literacy positively moderates the relationship between utilization of generative AI and its impact on R&D processes, such that the positive effect of utilization is stronger for employees with high AI literacy.*

Finally, improvements in R&D processes enable organizations to strengthen their innovation capacity. Employees with high AI literacy may better translate process improvements into innovative outputs because they are capable of identifying opportunities, integrating AI-driven insights, and collaborating effectively with AI. For employees with low AI literacy, even if R&D processes improve, the translation into innovation capacity may be weaker. Hence, AI literacy may strengthen the positive effect of R&D process improvements on innovation capacity. We postulate:

H5d. *AI literacy positively moderates the relationship between the impact of generative AI on R&D processes and innovation capacity, such that the positive effect of R&D improvements is stronger for employees with high AI literacy.*

Figure 3 illustrates the hypotheses described above and their interrelationships in the research framework.

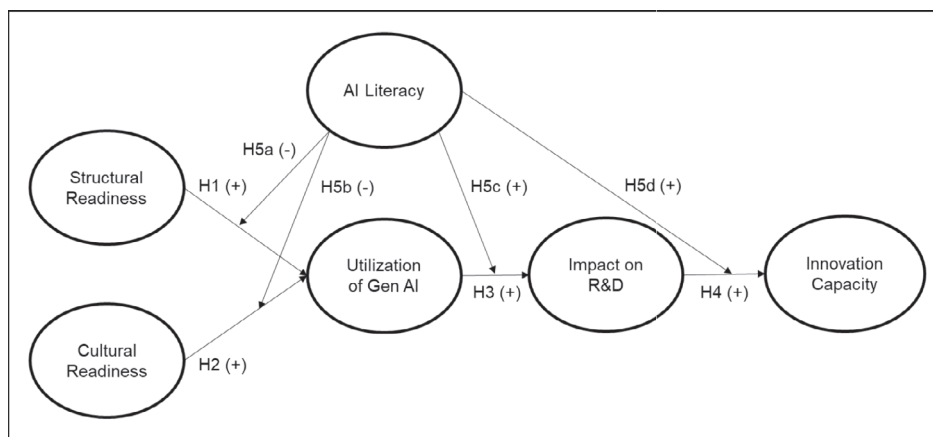


FIGURE 3 | Illustration of the research framework.

TABLE 3 | Overview over interviewees.

Interviewee	Country	Company	Position
B1	Germany	Large corporation	R&D manager
B2	Germany	Large corporation	Senior product developer
B3	Germany	Large corporation	Product developer
B4	Austria	Start-up	Head of product department
B5	Germany	Start-up	Founder of a software development company

4 | Method

In this study, we employed a mixed-method approach with an exploratory design (Creswell and Clark 2017) in sequential order (Tashakkori et al. 2020), as the qualitative method is used to help construct the quantitative instrument. The items are designed based on the results of the qualitative study. Accordingly, the questionnaire construction is supported by previous interviews. The qualitative pre-study was designed to explore the relevance of our research framework and crucial topics for the quantitative study as well as to evaluate the questionnaire and make adjustments if necessary.

4.1 | Qualitative Pre-Study

The selection of interviewees in this study is broadly diversified in order to obtain an assessment from various sectors. At the same time, care is taken to ensure that the interviewees meet the requirements of an expert. Table 3 presents an overview of our interviews. The semi-structured interviews were recorded and transcribed.

The interviews were analyzed with a qualitative content analysis, which is particularly advantageous, as it is well-suited for the mixed-methods research approach (Mayring and Fenzl 2014). The analysis process of inductive category formation is particularly suitable for the exploratory design. Once the material has been transcribed, the category definition is determined. When going through the transcripts for the first time, top categories are filtered out, such as the use of generative AI in the company surveyed and the future

development of the use of AI in companies, the positive impact on practice, parameters for measuring innovation capability, or the assessment of the questionnaire. The data material is then worked through again and further summarized. Reading through the transcripts several times prevents further categories from being forgotten or overlooked until the data has been exhausted and the researcher is of the opinion that no further categories can be found. This concept is also referred to as saturation (Barbour 2014).

4.2 | Quantitative Study

A structural equation model (SEM) is used to analyze the underlying complex relationships simultaneously. Six latent variables (structural readiness, cultural readiness, AI literacy, utilization of generative AI, impact on R&D, and innovation capability) are used in this paper. The online survey was created on the Lamapoll platform. The DACH region was selected as the target population, as the survey was conducted in German, which is predominantly spoken in Germany, Austria, and Switzerland. It also prevents the survey from having to be translated into another language, which in turn could distort the results. The survey was conducted from June 13, 2024, to August 1, 2024. A total of 291 individuals participated in the study.

The survey asks for general information such as gender, age, country, size and location of the company, industry, length of employment at the company, professional experience, level of education, and AI literacy of the respondent. Furthermore, the organizational readiness of companies for the use of generative AI is addressed. The designed questions are based on the already

TABLE 4 | Demographics of the sample ($N = 291$).

Category	Response	Percentage	Mean	Min.	Max.
Gender	Male	50.50	1.50	1.00	3.00
	Female	48.51			
	Diverse	0.99			
Age	Age in years		31.66	20.00	73.00
Education	Secondary school certificate (Hauptschulabschluss)	2.00	5.33	1.00	8.00
	Secondary school certificate (Realschulabschluss)	2.00			
	Vocational baccalaureate diploma (Fachabitur)	3.00			
	A-levels (Abitur)	7.50			
	Bachelor's degree	38.50			
	Master's degree	36.00			
	Diplom degree (Diplom-Abschluss)	8.50			
	Ph.D.	2.50			
Size (Number of employees)	< 9	7.46	3.18	1.00	4.00
	< 49	21.39			
	< 249	16.92			
	> 250	54.23			
Sector	Automotive	11.44	11.51	1.00	18.00
	Railroad	1.00			
	Education	4.48			
	Chemistry and pharmaceuticals	2.49			
	Electrical engineering and electronics	1.49			
	Energy supply	4.48			
	Precision mechanics	0.50			
	Healthcare	7.96			
	Information technology and telecommunications	14.93			
	Finance and insurance	4.98			
	Food	2.99			
	Aviation and aerospace	1.00			
	Machinery and plant engineering	6.47			
	Steel and metal	0.50			
	Textiles	1.99			
	Other	33.33			

existing and robust test instruments by the Texas Christian University (Institute of Behavioral Research 2011), Molla and Licker (2005), Stratman and Roth (2002), and Shahrabi and Paré (2014). Moreover, the questions relating to applications of generative AI in the area of R&D activities are based on publications by Brossard et al. (2020), Cooper and McCausland (2024), and the results of the expert interviews. Finally, the survey deals with the ability to innovate. These questions are based on indicators that are suitable for measuring innovation capability and

were formulated independently. These items are also critically scrutinized and adapted by means of the expert interviews. The demographics of the sample are presented in Table 4.

4.2.1 | Measures

Our research model encompasses six constructs (for measurement items, see Appendix S1).

4.2.1.1 | Structural Readiness. A five-point scale was used to operationalize the items: 1 = does not apply at all, to 5 = applies strongly. The variable consists of three items: (a) Company_Practices, (b) Tech_Tools, and (c) Data_Access. After conducting an exploratory and confirmatory factor analysis, the items (d) Use_Restrictions, (e) Trained_Staff, and (f) Address_Concerns were excluded.

4.2.1.2 | Cultural Readiness. This factor is measured with the following five items: (a) Openness_AI, (b) Clear_Goals, (c) Company_Values, (d) Data_Sensitivity, and (e) Company_Needs. These items were also asked on a five-point scale (1 = does not apply at all, to 5 = applies strongly).

4.2.1.3 | AI Literacy. In order to measure this factor, the items (a) Basic AI knowledge, (b) Algorithm Knowledge, and (c) Experience AI algorithms were used. Items (a) and (b) were operationalized with a five-point scale (1 = very poor, to 5 = very good). Item (c) was measured with a five-point scale ranging from 1 = I don't know what an AI algorithm is to 5 = I have developed AI algorithms myself.

4.2.1.4 | Utilization of Generative AI. This factor is operationalized using the five items (a) Use_Idea_Generation, (b) Use_Material_Research, (c) Use_Prototyping, (d) Use_Topology, and (e) Use_Product_Optimization. Here, too, the items are based on a five-point scale (1 = does not apply at all, to 5 = applies strongly).

4.2.1.5 | Impact on R&D Activities. In order to measure this factor, the items (a) Faster_Realization, (b)

Complex_Projects, (c) Visualize_Prototypes, (d) Product_Alternatives, and (e) Out_Of_The_Box were considered. A five-point scale was again used for operationalization: 1 = does not apply at all, to 5 = applies strongly. After the exploratory and confirmatory factor analysis, (e) Out_Of_The_Box was excluded.

4.2.1.6 | Innovation Capacity. The innovation capacity factor was operationalized using the items (a) Efficient_Development, (b) Faster_Development, and (c) More_Productinnovation. Here, too, the items are based on a five-point scale (1 = does not apply at all, to 5 = applies strongly). After the exploratory and confirmatory factor analysis, (d) Improved_Quality was excluded.

4.2.1.7 | Control Variables (CVs). We included size of the company (reference category: 1 (<9)), age of the respondents, and gender (reference category: 1 (male)) as CVs into our analysis. In addition, we included education, which we grouped into two categories, namely school-bachelor (reference category) and master-Ph.D. (see also Tables 4 and 7).

5 | Analysis

5.1 | Measurement Model

In the primary step, the latent constructs were tested to ensure an adequate level of reliability and validity of the measures. The factor loadings, Cronbach's alpha (α), composite reliability (CR), and the average variance extracted (AVE) are summarized in Table 5. The reliability of the measurements was evaluated using

TABLE 5 | Measurement results for latent constructs.

Factor	Item	Factor loadings	Cronbach's alpha	CR	AVE
Structural readiness	SR1	0.726	0.729	0.819	0.602
	SR2	0.846			
	SR3	0.750			
Cultural readiness	CR1	0.708	0.776	0.858	0.550
	CR2	0.853			
	CR3	0.756			
	CR4	0.644			
	CR5	0.730			
AI literacy	AIL1	0.890	0.865	0.917	0.787
	AIL2	0.872			
	AIL3	0.715			
Utilization of Gen AI	U1	0.703	0.835	0.887	0.612
	U2	0.754			
	U3	0.783			
	U4	0.866			
	U5	0.795			
Impact on R&D	I1	0.703	0.765	0.852	0.592
	I2	0.732			
	I3	0.843			
	I4	0.792			
Innovation capacity	IC1	0.816	0.770	0.870	0.690
	IC2	0.869			
	IC3	0.806			

composite reliability and Cronbach's alpha. As indicated in Table 5, the composite reliability of the latent constructs ranged from 0.819 to 0.917, exceeding the recommended threshold of 0.7, thereby demonstrating reliability (Chin 1998). Additionally, the reported Cronbach's alpha values, which range from 0.729 to 0.865, confirm satisfactory reliability. Nunnally and Bernstein (1994) suggest that a Cronbach's alpha of 0.7 or higher is deemed acceptable for research purposes. Convergent and discriminant validity are considered adequate when the AVEs of all constructs exceed 0.50 (Chin 1998; Hair Jr et al. 2021). As presented in Table 5, the AVEs range from 0.550 to 0.787, indicating that they surpass the required threshold. The discriminant validity of the constructs is evaluated using the Fornell-Larcker criterion. Consistent with recommended practice, all latent constructs exhibit square root values of the AVE that are greater than their intercorrelations with other corresponding constructs (see Table 6; diagonal values exceed all other values in their respective rows). These results indicate adequate discriminant validity (Chin 1998, Hair Jr et al. 2021).

5.2 | Structural Model

We used SEM in SmartPLS to test our hypotheses.² In order to determine an appropriate sample size, we conducted a power analysis (e.g., Moshagen and Bader 2024), which resulted in a recommended minimum sample size for effect detection of 100. Thus, the results can be considered meaningful.

The relationship between structural readiness and utilization of generative AI is not significant ($\beta=0.132$; $p=0.244$). Therefore, hypothesis 1 is not supported.

The relationship between cultural readiness and utilization of generative AI has a positive and significant influence ($\beta=0.289$; $p<0.01$), which leads to the conclusion that cultural readiness for AI has a positive influence on the use of generative AI. This supports hypothesis 2.

With regard to the impact on R&D activities, the use of generative AI has a positive influence on R&D activities, as the link is also significant ($\beta=0.493$; $p<0.001$). As a result, hypothesis 3 is also supported.

Moreover, the model confirms the influence of the impact of R&D activities supported by generative AI on the innovation capacity,

that is, the impact significantly influences the ability to innovate ($\beta=0.413$; $p<0.001$). Therefore, hypothesis 4 is supported.

AI literacy significantly moderates the relationship between structural readiness and utilization of generative AI ($\beta=-0.297$; $p<0.01$). However, the moderation effect of the relationship between cultural readiness and utilization of generative AI is not significant ($\beta=0.094$; $p=0.241$). Thus, hypothesis 5a is supported, but not hypothesis 5b.

Finally, AI literacy has a significant negative moderation effect on the relationship between the utilization of generative AI and its impact on R&D activities ($\beta=-0.135$; $p<0.05$), but not on the relationship between the impact on R&D activities and innovation capacity ($\beta=0.045$; $p=0.698$). As a result, hypotheses 5c and 5d are not supported.³

A PLS bootstrapping procedure with 5000 subsamples (Hair Jr et al. 2021) was conducted to assess the statistical significance and relevance of the structural model's path coefficients.

Figure 4 shows the graphical representations of the results, including path coefficients and their statistical significance, and Tables 7 and 8 summarize the results of our hypothesis testing.

In order to better understand the significant moderating effects of AI literacy, we conducted a simple slope analysis. The simple slope diagram for the moderation of the relationship between structural readiness and utilization of generative AI (Figure 5) indicates that at -1 SD (low AI literacy), the plot is steep with a positive gradient compared to $+1$ SD (high AI literacy), where the gradient is negative. This means that for employees with low AI literacy, an increase in structural readiness enhances their utilization of generative AI. Conversely, for employees with high AI literacy, an increase in structural readiness reduces their utilization of generative AI. Thus, as employees' AI literacy increases, the positive relationship between structural readiness and utilization of generative AI weakens and eventually even becomes negative. For the moderation of the relationship between utilization of generative AI and its impact on R&D processes, the simple slope diagram (Figure 6) shows that at -1 SD (low AI literacy), the plot is steeper with a positive gradient compared to $+1$ SD (high AI literacy). Thus, as employees' AI literacy increases, the relationship between utilization of generative AI and its impact on R&D processes weakens.

TABLE 6 | Discriminant validity and correlation.

	Structural readiness	Cultural readiness	Utilization of Gen AI	Impact on R&D	Innovation capability	AI literacy
Structural readiness	0.805					
Cultural readiness	0.591	0.735				
Utilization of Gen AI	0.254	0.422	0.777			
Impact on R&D	0.278	0.264	0.462	0.764		
Innovation capability	0.060	0.267	0.306	0.410	0.829	
AI literacy	0.188	0.278	0.161	0.146	0.031	0.884

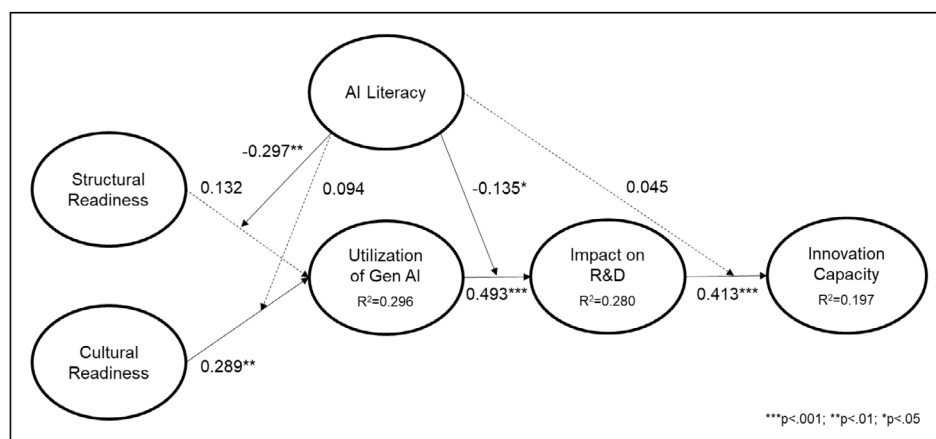


FIGURE 4 | Results of the structural model (standardized coefficients).

6 | Discussion

Our study investigates how organizational readiness affects the use of generative AI, how generative AI in turn impacts R&D activities and the organizations' innovation capacity, and how these effects are moderated by employees' AI literacy. Structural readiness for AI has no significant impact on the use of generative AI. This means that, according to these results, AI-related work practices, technical equipment, and access to the necessary data sets had no direct impact on the use of generative AI in companies. However, the effect of structural readiness is significantly moderated by employees' AI literacy ($\beta = -0.297$). Our results indicate that AI literacy negatively moderates the relationship between structural readiness and the utilization of generative AI. Specifically, structural readiness has a positive effect on utilization when employees possess low AI literacy but turns negative when AI literacy is high. This suggests that organizational structures can act as substitutes for limited individual knowledge, providing guidance and lowering barriers for employees with low AI literacy. However, for employees with high AI literacy, the same structures may be perceived as overly rigid or misaligned with their advanced competencies. In these cases, existing structures may hinder experimentation and reduce motivation to engage with generative AI tools. From the perspective of absorptive capacity, AI literacy represents a microfoundation that enables employees to acquire, assimilate, transform, and exploit new knowledge. When AI literacy is low, structural readiness strengthens absorptive capacity by compensating for individual-level limitations. Conversely, when AI literacy is high, structural mechanisms can restrict the transformation and exploitation of knowledge, thereby dampening realized absorptive capacity. This finding highlights that absorptive capacity as a dynamic capability is not simply strengthened by the accumulation of both organizational and individual microfoundations, but rather depends on the alignment between them. Structural readiness therefore plays a dual role: it enables utilization in contexts of low or medium AI literacy but may discourage utilization or constrain exploratory use in contexts of high AI literacy.

Cultural readiness has a significant positive effect on the utilization of generative AI ($\beta = 0.289$). Thus, our findings emphasize the importance of an open culture with regard to the implementation of AI in a company. Cultural readiness that encompasses

clear implementation goals, AI-compatible corporate values, corporate needs, and suitable training measures is of immense importance for the implementation of generative AI. Since the moderation of AI literacy is not significant, the effect of cultural readiness is equally important for any level of AI literacy.

Once generative AI is used in the organization, it has a significant positive impact on R&D activities ($\beta = 0.493$). This means that generative AI will contribute to idea generation, material research, prototyping, topology, or product optimization. These activities are likely to benefit significantly from the utilization of generative AI, as it facilitates the rapid implementation of specific product requirements while reducing the need for multiple prototypes to visualize them. Additionally, generative AI enables employees without prior expertise to contribute to larger and more complex projects, leading to the generation of a broader range of product alternatives. Consequently, a substantial reduction in development process timelines can be anticipated, enhancing overall efficiency and productivity. Interestingly, the positive impact of generative AI utilization on R&D activities is significantly moderated by AI literacy ($\beta = -0.135$) such that the effect is weaker the higher the AI literacy of an employee. While utilization generally enhances R&D outcomes by streamlining prototyping, expanding problem-solving, and improving idea generation, this positive effect diminishes as employees' AI literacy increases. One explanation is that employees with higher AI literacy may set more ambitious or sophisticated expectations for AI, making the incremental benefits of utilization appear less pronounced compared to those with lower literacy, who experience more immediate efficiency and creativity gains. Another possible mechanism is that highly AI-literate employees tend to use AI more critically, questioning its outputs and integrating them cautiously into R&D workflows (Fügener et al. 2022; Ng et al. 2021), which can reduce the apparent short-term impact. From the perspective of absorptive capacity, AI literacy equips employees with the skills to acquire and assimilate knowledge, but it may also lead to a more selective transformation and exploitation process. Thus, while high AI literacy enhances potential absorptive capacity, it may simultaneously constrain realized absorptive capacity by fostering critical filtering and a reliance on complementary human expertise. This pattern underscores that the benefits of generative AI in R&D are not uniform but contingent on the interplay between utilization and employees' knowledge foundations.

TABLE 7 | Overview of results.

	Path	β	<i>t</i> statistics	<i>p</i>	Significance
H1	SR → U	0.132	1.165	0.244	
H2	CR → U	0.289	3.068	0.003	**
H3	U → I	0.493	6.848	0.000	***
H4	I → IC	0.413	4.362	0.000	***
H5a	AIL × SR → U	−0.297	3.190	0.001	**
H5b	AIL × CR → U	0.094	1.173	0.241	
H5c	AIL × U → I	−0.135	2.207	0.027	*
H5d	AIL × I → IC	0.045	0.388	0.698	
	AIL → U	0.086	1.230	0.219	
	AIL → I	0.102	1.381	0.167	
	AIL → IC	−0.043	0.475	0.635	
CV	Size_2 → U	−0.270	0.888	0.374	
	Size_2 → I	0.058	0.210	0.834	
	Size_2 → IC	0.295	1.043	0.297	
	Size_3 → U	−0.166	0.550	0.582	
	Size_3 → I	0.018	0.064	0.949	
	Size_3 → IC	0.370	1.281	0.200	
	Size_4 → U	−0.269	0.935	0.350	
	Size_4 → I	−0.072	0.264	0.791	
	Size_4 → IC	0.333	1.202	0.230	
	Age → U	0.021	0.424	0.672	
	Age → I	0.102	1.381	0.167	
	Age → IC	0.074	1.067	0.286	
	Gender_2 → U	0.093	0.599	0.549	
	Gender_2 → I	0.366	2.529	0.011	*
	Gender_2 → IC	0.031	0.204	0.839	
	Gender_3 → U	0.504	0.161	0.872	
	Gender_3 → I	−0.364	0.117	0.907	
	Gender_3 → IC	0.404	0.144	0.885	
	Edu_postgrad → U	0.091	0.572	0.567	
	Edu_postgrad → I	−0.081	0.518	0.605	
	Edu_postgrad → IC	0.288	1.780	0.075	

Note: Path significances: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Abbreviations: AIL, AI literacy; CR, cultural readiness; I, impact on R&D processes; IC, innovation capacity; SR, structural readiness; U, utilization of generative AI.

The positive effect of generative AI on R&D activities in turn has a significantly positive effect on the organization's innovation capacity ($\beta = 0.413$). This is reflected in more efficient design processes and accelerated development timelines, enabling the creation of a greater number of products within a shorter period. This effect is not moderated by employees' AI literacy.

6.1 | Theoretical Contributions

This study makes several significant theoretical contributions, advancing the existing literature on organizational readiness, AI literacy, generative AI, human–AI collaboration, R&D activities, and innovation capacity. First, it enriches the body

TABLE 8 | Overview of hypotheses.

	Hypothesis	Result
H1	The more the structural readiness is geared towards the introduction of AI, the more likely it is that generative AI will be used in the company	Not supported
H2	The more the cultural readiness is geared towards the introduction of AI, the more likely it is that generative AI will be used in the company	Supported
H3	The utilization of generative AI has a positive impact on research and development activities	Supported
H4	Research and development activities that are enhanced by generative AI have a positive impact on the innovation capacity of a company	Supported
H5a	AI literacy negatively moderates the relationship between structural readiness and utilization of generative AI, such that the positive effect of structural readiness is stronger for employees with low AI literacy and weaker for employees with high AI literacy	Supported
H5b	AI literacy negatively moderates the relationship between cultural readiness and utilization of generative AI, such that the positive effect of cultural readiness is stronger for employees with low AI literacy and weaker for employees with high AI literacy	Not supported
H5c	AI literacy positively moderates the relationship between utilization of generative AI and its impact on R&D processes, such that the positive effect of utilization is stronger for employees with high AI literacy	Not supported (significantly negative)
H5d	AI literacy positively moderates the relationship between the impact of generative AI on R&D processes and innovation capacity, such that the positive effect of R&D improvements is stronger for employees with high AI literacy	Not supported

of research on organizational readiness by addressing its role in the context of AI usage. While much of the existing literature has focused on readiness for traditional IT systems (e.g., Shahrasbi and Paré 2015), this study investigates two distinct dimensions of readiness—structural and cultural—in the context of generative AI. The findings suggest that readiness is technology-contingent, functioning differently for generative AI (which is interactive, creative, and probabilistic) than for more deterministic IT systems. Notably, the findings suggest that cultural readiness plays a more pivotal role in facilitating the utilization of generative AI, as it fosters shared values and a positive mindset toward adoption across the organization regardless of the level of employees' AI literacy. In contrast, the effect of structural readiness is starkly opposing for different levels of AI literacy. Our findings demonstrate that structural and cultural readiness do not simply add up; rather, they operate through different psychological and operational pathways. These insights provide a nuanced understanding of how different aspects of organizational readiness interact with emerging AI technologies, thereby extending theoretical frameworks on organizational readiness as a microfoundation of absorptive capacity as a dynamic capability.

Second, this paper integrates AI literacy, which has primarily been studied in IS research, into R&D and innovation management research. In doing so, this study contributes to theory by demonstrating that AI literacy, as a microfoundation of absorptive capacity, does not uniformly strengthen the relationship between organizational readiness, utilization of generative AI, and its impact on R&D. Our findings extend literature by highlighting the importance of alignment between organizational and individual-level microfoundations. They suggest that absorptive capacity is not simply cumulative,

with organizational readiness and individual AI literacy always reinforcing one another. Instead, structural readiness can either enable or hinder utilization depending on employee AI literacy levels. Moreover, the impact of utilization may exhibit diminishing marginal returns and depend on how critically AI outputs are processed. Theoretically, this points to the need for a more nuanced understanding of microfoundations, suggesting that dynamic capabilities depend not only on their presence but also on their interaction across organizational and individual levels.

Third, this study deepens the still limited understanding of how employees with different levels of AI literacy engage with generative AI in general and in R&D in particular. Prior research has only begun to explore how varying levels of familiarity, skill, and confidence with AI shape actual usage behaviors and outcomes. By empirically examining how user groups differ in their utilization of generative AI and how this shapes their work outcomes, we show that AI literacy does not simply increase usage but meaningfully conditions how generative AI is incorporated into daily work. These insights help explain why employees may realize different benefits or even drawbacks from the same technological tools. Therefore, our work advances an emerging but still underdeveloped research area on the heterogeneous nature of generative AI use in organizational settings (e.g., Colville and Ostern 2024).

In doing so, this research also advances the literature on human–AI collaboration in organizational settings (e.g., Hou et al. 2025; Chen and Chan 2024) by examining how individual attributes, particularly AI literacy, shape employees' collaboration with generative AI and influence the outcomes of their work processes. Our findings indicate that AI literacy shapes

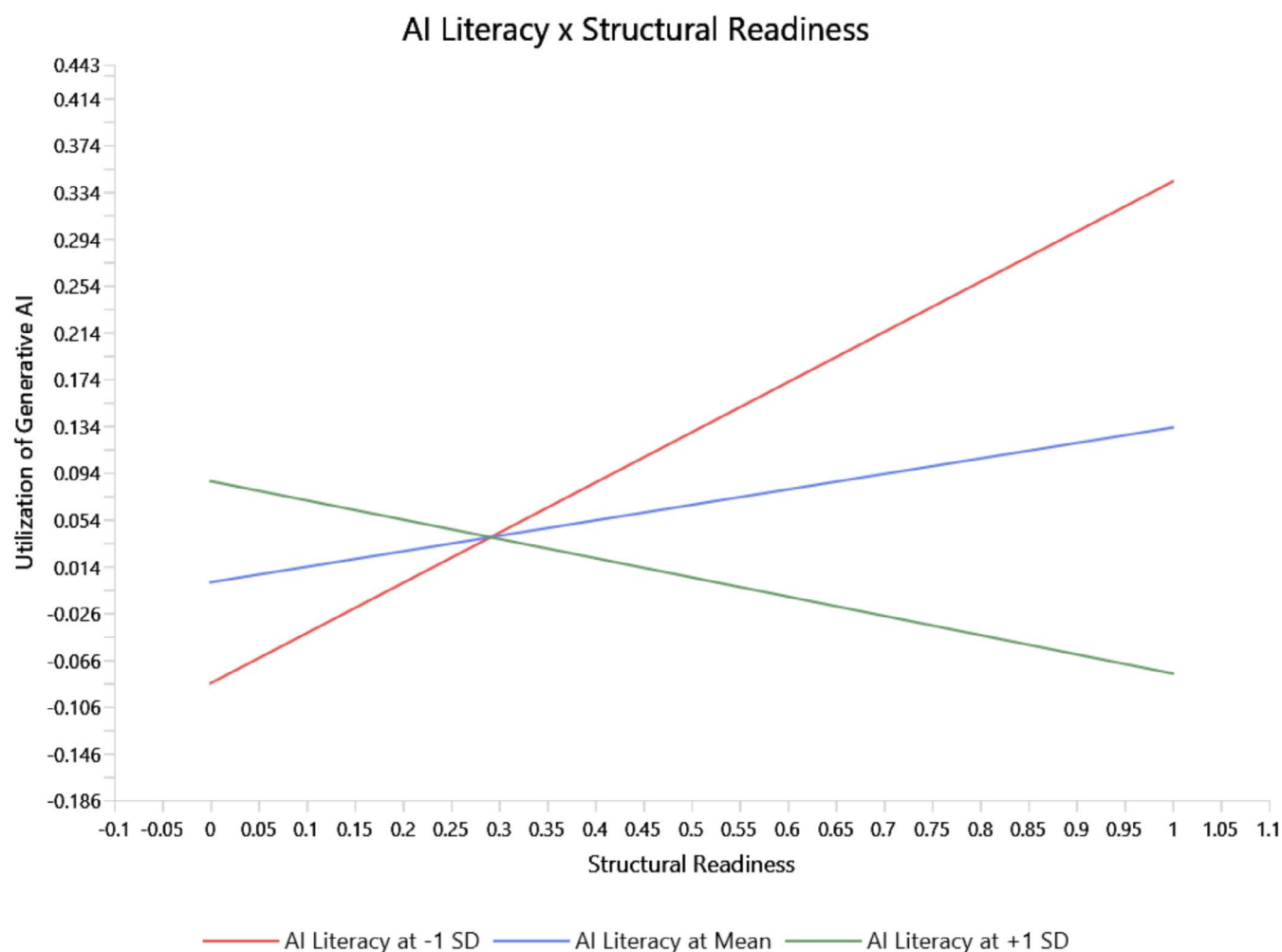


FIGURE 5 | Moderating effect of AI literacy on structural readiness and utilization of generative AI.

not only whether employees use generative AI, but how they integrate it into their tasks, evaluate its output, and balance human judgment with algorithmic input. By uncovering these mechanisms, we shed light on the nuanced and often complex ways in which R&D outcomes emerge when humans and generative AI jointly contribute to problem-solving and knowledge creation.

Fourth, this paper provides empirical evidence that generative AI can enhance R&D activities within organizations. By demonstrating how generative AI supports idea generation, material research, prototyping, topology optimization, and creative problem-solving, the study highlights its potential to transform traditional R&D practices. Beyond documenting performance-related benefits, our findings also show how generative AI affects R&D processes—for example, by enabling faster implementation of product or service requirements, broadening employee participation in complex development tasks, facilitating the visualization of concepts with fewer physical prototypes, and expanding the range of viable product or service alternatives. In doing so, the study moves beyond conceptual claims and offers empirical examinations of generative AI's role in the micro-processes of organizational R&D. These insights contribute to the growing literature on AI in corporate settings by substantiating the concrete

mechanisms through which generative AI affects innovation-related activities and by demonstrating that its value extends beyond automation to the augmentation of human creativity and experimentation in R&D contexts.

Finally, the study contributes to the research on organizational innovation capacity by unraveling the pathways through which generative AI impacts this critical organizational capability. Specifically, it illustrates how the integration of generative AI in R&D can enhance design processes, accelerate development timelines, and enable the creation of a greater number of products. This connection between generative AI, R&D activities, and innovation capacity adds a novel perspective to the discourse on how emerging technologies shape organizational competitiveness and adaptability in dynamic markets.

6.2 | Practical Implications

This study provides several practical implications for organizations aiming to integrate generative AI into R&D. First, our findings emphasize the importance of cultural readiness. Managers should actively cultivate a supportive AI culture by clearly communicating its benefits, aligning initiatives with organizational values, and building employee confidence in working alongside

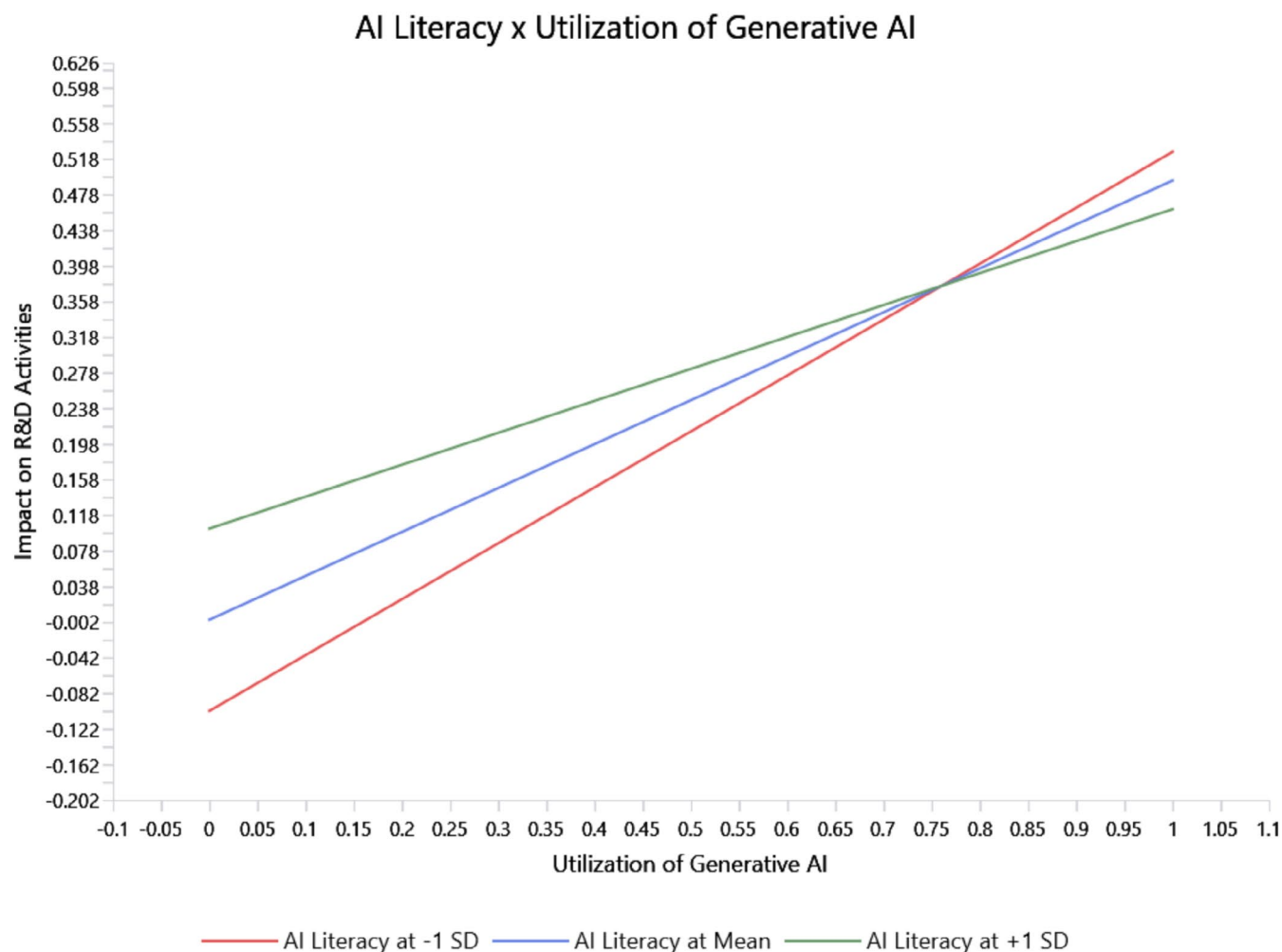


FIGURE 6 | Moderating effect of AI literacy on utilization of generative AI and impact of generative AI on R&D activities.

AI. Such a cultural foundation encourages acceptance, collaboration, and proactive engagement with generative AI.

Second, the moderation effect of AI literacy on structural readiness shows that uniform strategies are ineffective. For employees with low AI literacy, strong structural support such as compatible work practices or adequate equipment lowers adoption barriers. By contrast, for highly literate employees, rigid structures may discourage use, making greater autonomy, flexibility, and experimental spaces more effective. Organizations should therefore design differentiated approaches that match readiness strategies to employee skill levels.

Third, the study demonstrates that generative AI enhances R&D by shortening development timelines, expanding ideation, and supporting rapid prototyping and experimentation. These capabilities enable firms to increase efficiency and productivity while involving employees with diverse skills in complex innovation projects. At the same time, managers should recognize that highly AI-literate employees may apply AI more critically, reducing immediate performance gains compared to less literate colleagues. Firms can address this by pairing critical evaluators with rapid implementers or embedding structured reflection practices that balance quality with speed.

Finally, the results suggest that generative AI should be positioned strategically as a driver of competitive advantage. Achieving this requires ongoing investments in training, process refinement, and cross-functional collaboration to embed generative AI effectively into innovation ecosystems. Policymakers also play an important role by funding literacy programs, incentivizing readiness investments, and establishing clear regulatory frameworks on responsible AI use. Together, these efforts can ensure that generative AI becomes a sustainable enabler of innovation capacity.

6.3 | Limitations and Future Research

While this study provides valuable insights into the organizational readiness, AI literacy, and the impact of generative AI on R&D activities and innovation capacity, it is not without limitations. These limitations highlight opportunities for future research. First, the scope of the study was constrained to specific dimensions of organizational readiness, namely cultural and structural readiness. While these dimensions are central to understanding the utilization of generative AI, future research could explore additional facets of readiness, such as financial readiness or business readiness (Shahrabi and Paré 2014) to provide a more holistic view of organizational preparedness for AI integration.

Second, given the still low rates of AI utilization, our sample size was relatively small, although it included companies across sectors. Future large-scale research could conduct industry-specific studies to examine potential differences among sectors.

Third, AI utilization remains limited in general and was found to be particularly low in the very early stages of R&D processes in our pre-study—findings consistent with those of Cooper and Brem (2024). Consequently, we were unable to focus on this aspect, as our investigation depended on the actual usage of generative AI. Future studies could address this shortcoming by including the very early stages of the R&D process, with a focus on activities such as knowledge search (e.g., Sofka and Grimpe 2010), where generative AI has the potential to support efficient and comprehensive information retrieval and processing.

Finally, future research could draw on affordance theory to examine how generative AI's possibilities are interpreted and enacted in practice, and on institutional theory to explore how industries, organizations, or teams may begin to mimic one another as AI-related routines and norms diffuse.

7 | Conclusion

This study provides important insights into the organizational readiness for generative AI, AI literacy, and the impact of generative AI on R&D activities and innovation capacity. It advances understanding of generative AI in R&D by moving beyond questions of adoption and efficiency to emphasize its strategic implications for innovation management. Rather than functioning merely as a productivity-enhancing tool, generative AI emerges as a capability-building resource whose value depends critically on organizational readiness and individual AI literacy. Our findings underscore that cultural readiness, i.e., shared values, openness, and a supportive mindset toward AI, constitutes a foundational condition for meaningful utilization, while structural readiness exerts differentiated effects depending on employees' AI literacy. Together, these insights highlight that the strategic payoff of generative AI lies not in technological deployment alone, but in how organizations cultivate the conditions under which AI can be productively integrated into R&D work.

From an R&D management perspective, these results call for a rethinking of how innovation activities are organized and governed in the AI era. Generative AI reshapes core R&D micro-processes by accelerating experimentation, broadening participation in complex development tasks, and augmenting creative and analytical work. Managing R&D in this context therefore requires a shift from optimizing isolated tools toward orchestrating human–AI collaboration, aligning organizational structures with employee capabilities, and deliberately designing learning and experimentation spaces. In this sense, the introduction of generative AI represents not an incremental technological upgrade but a managerial transition comparable to earlier shifts associated with digitalization and advanced automation.

This study points to the broader potential of AI-readiness frameworks to reshape innovation management research and

practice. By conceptualizing organizational readiness and AI literacy as interacting microfoundations of absorptive capacity, our findings suggest that future theories of innovation should account more explicitly for the co-evolution of technologies, organizational conditions, and human capabilities. Looking ahead, integrating AI-readiness perspectives with established innovation theories offers a promising avenue for understanding how organizations build, sustain, and renew innovation capabilities in increasingly AI-intensive environments. As generative AI continues to evolve, the challenge for both scholars and practitioners will be not simply to keep pace with technological change, but to rethink how innovation itself is organized, enabled, and governed.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy restrictions.

Endnotes

¹ The term “product” encompasses both tangible and intangible outputs in a modern context as products and services are complementing each other in various industries and product-service systems are pervasive (e.g., Shelton 2009; Alfarisi et al. 2024).

² SmartPLS estimates latent constructs and path coefficients using the PLS-SEM algorithm (Wold 1982; Lohmöller 1989; Ringle et al. 2024).

³ To examine all four moderation effects more comprehensively, we also tested for quadratic (curvilinear) effects. These were, however, not statistically significant. Hence, the relationships are linear.

References

- Adhiatma, A., O. Fachrunnisa, Nurhidayati, and T. Rahayu. 2022. “Creating Digital Ecosystem for Small and Medium Enterprises: The Role of Dynamic Capability, Agile Leadership and Change Readiness.” *Journal of Science and Technology Policy Management* 14: 941–959.
- Aggarwal, A., M. Mittal, and G. Battineni. 2021. “Generative Adversarial Network: An Overview of Theory and Applications.” *International Journal of Information Management Data Insights* 1: 100004.
- Alfarisi, S., Y. Shimomura, and I. Masudin. 2024. “Advancing Product Service Systems - Life Cycle Assessment: Robust Method for Sustainability Assessment.” *Cleaner Production Letters* 7: 100081.
- Barbour, R. 2014. *Introducing Qualitative Research: A Student's Guide*. Second Edition ed. SAGE Publications, Inc.
- Bilgram, V., D. Cañadas Link, and C. Lang-Koetz. 2023. *Generative KIs in Kreativprozessen: Praxiserfahrungen Aus den ersten Monaten Mit ChatGPT & Co. Ideen-Und Innovationsmanagement*.

- Bilgram, V., and F. Laarmann. 2023. "Accelerating Innovation With Generative AI: AI-Augmented Digital Prototyping and Innovation Methods." *IEEE Engineering Management Review* 51: 18–25.
- Bouschery, S. G., V. Blazevic, and F. T. Pillier. 2023. "Augmenting Human Innovation Teams With Artificial Intelligence: Exploring Transformer-Based Language Models." *Journal of Product Innovation Management* 40: 139–153.
- Brem, A., F. Giones, and M. Werle. 2023. "The AI Digital Revolution in Innovation: A Conceptual Framework of Artificial Intelligence Technologies for the Management of Innovation." *IEEE Transactions on Engineering Management* 70: 770–776.
- Brossard, M., G. Gatto, A. Gentile, T. Merle, and C. Wlezien. 2020. *How Generative Design Could Reshape the Future of Product Development*. McKinsey & Company.
- Brynjolfsson, E., and A. McAfee. 2017. "The Business of Artificial Intelligence: What It Can—And Cannot—Do for Your Organization." *Harvard Business Review* 7: 3–11.
- Cetindamar, D., K. Kitto, M. Wu, Y. Zhang, B. Abedin, and S. Knight. 2024. "Explicating AI Literacy of Employees at Digital Workplaces." *IEEE Transactions on Engineering Management* 71: 810–823.
- Chen, Z., and J. Chan. 2024. "Large Language Model in Creative Work: The Role of Collaboration Modality and User Expertise." *Management Science* 70: 9101–9117.
- Chin, W. W. 1998. "The Partial Least Squares Approach to Structural Equation Modeling." *Modern Methods for Business Research*.
- Coch, L., and J. R. P. French. 1948. "Overcoming Resistance to Change." *Human Relations* 1: 512–532.
- Cohen, W. M., and D. A. Levinthal. 1990. "Absorptive Capacity: A New Perspective on Learning and Innovation." *Administrative Science Quarterly* 35: 128–152.
- Colville, S., and N. Ostern. 2024. *Trust and Distrust in GAI Applications: The Role of AI Literacy and Metaknowledge. Proceedings of the International Conference on Information Systems (ICIS 2024)*. Association for Information Systems.
- Cooper, R. G. 2024. "The Artificial Intelligence Revolution in New-Product Development." *IEEE Engineering Management Review* 52: 195–211.
- Cooper, R. G., and A. M. Brem. 2024. "The Adoption of AI in New Product Development." *Research-Technology Management* 67: 44–53.
- Cooper, R. G., and T. McCausland. 2024. "AI and New Product Development." *Research-Technology Management* 67: 70–75.
- Creswell, J. W., and V. L. P. Clark. 2017. *Designing and Conducting Mixed Methods Research*. London, New Delhi, Singapore, Sage publications.
- Davis, F. D., R. P. Bagozzi, and P. R. Warshaw. 1989. "User Acceptance of Computer Technology: A Comparison of Two Theoretical Models." *Management Science* 35: 982–1003.
- De Ridder, P., N. Bogaert, and A. Co-Pilot. 2023. "The Age of Creative AI." <https://www.boardofinnovation.com/creative-ai/>.
- Dell'Acqua, F., E. Mcfowland, E. R. Mollick, et al. 2023. *Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality*. Harvard Business School Technology & Operations Mgt. Unit Working Paper, 1–58. Harvard Business School.
- Deng, J., and Y. Lin. 2022. "The Benefits and Challenges of ChatGPT: An Overview." *Frontiers in Computing and Intelligent Systems* 2: 81–83.
- Deszca, G., C. Ingols, and T. F. Cawsey. 2020. *Organizational Change: An Action-Oriented Toolkit*. London, New Delhi, Singapore, Washington, Dc, Melbourne, Sage Publications.
- Dwivedi, Y. K., N. Kshetri, L. Hughes, et al. 2023. "Opinion Paper: 'So What if ChatGPT Wrote It?' Multidisciplinary Perspectives on Opportunities, Challenges and Implications of Generative Conversational AI for Research, Practice and Policy." *International Journal of Information Management* 71: 102642.
- Eby, L. T., D. M. Adams, J. E. A. Russell, and S. H. Gaby. 2000. "Perceptions of Organizational Readiness for Change: Factors Related to Employees' Reactions to the Implementation of Team-Based Selling." *Human Relations* 53: 419–442.
- Eisenhardt, K. M., N. R. Furr, and C. B. Bingham. 2010. "CROSSROADS—Microfoundations of Performance: Balancing Efficiency and Flexibility in Dynamic Environments." *Organization Science* 21: 1263–1273.
- Felin, T., N. J. Foss, and R. E. Ployhart. 2015. "The Microfoundations Movement in Strategy and Organization Theory." *Academy of Management Annals* 9: 575–632.
- Feuerriegel, S., J. Hartmann, C. Janiesch, and P. Zschech. 2024. "Generative AI." *Business & Information Systems Engineering* 66: 111–126.
- Forsman, H. 2011. "Innovation Capacity and Innovation Development in Small Enterprises. A Comparison Between the Manufacturing and Service Sectors." *Research Policy* 40: 739–750.
- Fügener, A., J. Grahl, A. Gupta, and W. Ketter. 2022. "Cognitive Challenges in Human–Artificial Intelligence Collaboration: Investigating the Path Toward Productive Delegation." *Information Systems Research* 33: 678–696.
- Füller, J., K. Hutter, J. Wahl, V. Bilgram, and Z. Tekic. 2022. "How AI Revolutionizes Innovation Management – Perceptions and Implementation Preferences of AI-Based Innovators." *Technological Forecasting and Social Change* 178: 121598.
- Garbuio, M., and N. Lin. 2021. "Innovative Idea Generation in Problem Finding: Abductive Reasoning, Cognitive Impediments, and the Promise of Artificial Intelligence." *Journal of Product Innovation Management* 38: 701–725.
- Giordano, V., I. Spada, F. Chiarello, and G. Fantoni. 2024. "The Impact of ChatGPT on Human Skills: A Quantitative Study on Twitter Data." *Technological Forecasting and Social Change* 203: 123389.
- Gräfe, H., and S. Kauffeld. 2024. "ORC-Q: Toward the Multidimensional Measurement of Organizational Readiness for Change." *Diagnostica* 70: 77–87.
- Hair, J. F., Jr., T. G. Hult, C. M. Ringle, and M. Sarstedt. 2021. *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)*. Sage Publications.
- Holt, D. T., A. A. Armenakis, H. S. Feild, and S. G. Harris. 2007. "Readiness for Organizational Change: The Systematic Development of a Scale." *Journal of Applied Behavioral Science* 43: 232–255.
- Hou, J., L. Wang, G. Wang, H. J. Wang, and S. Yang. 2025. "The Double-Edged Roles of Generative AI in the Creative Process: Experiments on Design Work." *Information Systems Research*, 1–20. <https://doi.org/10.1287/isre.2024.0937>.
- Hurtado-Palomino, A., B. De La Gala-Velásquez, and J. Ccorisapra-Quintana. 2022. "The Interactive Effect of Innovation Capability and Potential Absorptive Capacity on Innovation Performance." *Journal of Innovation & Knowledge* 7: 100259.
- Iacovou, C. L., I. Benbasat, and A. S. Dexter. 1995. "Electronic Data Interchange and Small Organizations: Adoption and Impact of Technology." *MIS Quarterly* 19: 465–485.
- Institute of Behavioral Research. 2011. *TCU ORC-D4 (4-Domain Assessments for Organizational Readiness for Change)*. Texas Christian University.
- Johnson, P. C., C. Laurell, M. Ots, and C. Sandström. 2022. "Digital Innovation and the Effects of Artificial Intelligence on Firms' Research and Development – Automation or Augmentation, Exploration or Exploitation?" *Technological Forecasting and Social Change* 179: 121636.

- Jones, R. A., N. L. Jimmieson, and A. Griffiths. 2005. "The Impact of Organizational Culture and Reshaping Capabilities on Change Implementation Success: The Mediating Role of Readiness for Change." *Journal of Management Studies* 42: 361–386.
- Joosten, J., V. Bilgram, A. Hahn, and D. Totzek. 2024. "Comparing the Ideation Quality of Humans With Generative Artificial Intelligence." *IEEE Engineering Management Review* 52: 153–164.
- Just, J. 2024. "Natural Language Processing for Innovation Search – Reviewing an Emerging Non-Human Innovation Intermediary." *Technovation* 129: 102883.
- Kakatkar, C., V. Bilgram, and J. Füller. 2020. "Innovation Analytics: Leveraging Artificial Intelligence in the Innovation Process." *Business Horizons* 63: 171–181.
- Krishnan, V., and K. T. Ulrich. 2001. "Product Development Decisions: A Review of the Literature." *Management Science* 47: 1–21.
- Kulkarni, A., A. Shivananda, A. Kulkarni, and D. Gudivada. 2023. *Applied Generative AI for Beginners: Practical Knowledge on Diffusion Models, Chatgpt, and Other LLMs*. Ca, Apress.
- Lewin, K. 1947. "Frontiers in Group Dynamics:II. Channels of Group Life; Social Planning and Action Research." *Human Relations* 1: 143–153.
- Lewin, K., and D. Cartwright. 1964. *Field Theory in Social Science: Selected Theoretical Papers*. Harper & Row.
- Lohmöller, J.-B. 1989. *Latent Variable Path Modeling With Partial Least Squares*. Physica.
- Long, D., and B. Magerko. 2020. "What Is AI Literacy? Competencies and Design Considerations." In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery.
- Mariani, M., and Y. K. Dwivedi. 2024. "Generative Artificial Intelligence in Innovation Management: A Preview of Future Research Developments." *Journal of Business Research* 175: 114542.
- Mayring, P., and T. Fenzl. 2014. "Qualitative Inhaltsanalyse." In *Handbuch Methoden der Empirischen Sozialforschung*, edited by N. Baur and J. Blasius. Springer Fachmedien Wiesbaden.
- Mittal, N., C. Perricos, K. Schmidt, B. Sniderman, and D. Jarvis. 2024. *Now Decides Next: Getting Real About Generative AI in Germany*. Deloitte Center for Integrated Research.
- Molla, A., and P. S. Licker. 2005. "eCommerce Adoption in Developing Countries: A Model and Instrument." *Information & Management* 42: 877–899.
- Montasser, D., R. Prijadi, and T. E. Balqiah. 2023. "The Mediating Effect of IT-Enabled Dynamic Capabilities and Organizational Readiness on the Relationship Between Transformational Leadership and Digital Business Model Innovation: Evidence From Indonesia Incumbent Firms." *SAGE Open* 13: 21582440231181588.
- Moshagen, M., and M. Bader. 2024. "semPower: General Power Analysis for Structural Equation Models." *Behavior Research Methods* 56: 2901–2922.
- Müller, A. 2022. "BASF Sets a New Standard for Efficient Product Development of Sustainable Cosmetics With the Launch of Emollient Maestro." <https://www.basf.com/global/en/media/news-releases/2022/02/p-22-144>.
- Murphy, K. P. 2012. *Machine Learning: A Probabilistic Perspective*. London, MIT press.
- Narain, K., P. Daugherty, K. Schelfaut, D. Wood, P. P. Shukla, and J. Wilson. 2024. *Reinventing With a Digital Core*. Accenture.
- Ng, D. T. K., J. K. L. Leung, S. K. W. Chu, and M. S. Qiao. 2021. "Conceptualizing AI Literacy: An Exploratory Review." *Computers and Education: Artificial Intelligence* 2: 100041.
- Noy, S., and W. Zhang. 2023. "Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence." *Science* 381: 187–192.
- Nunnally, J. C., and I. H. Bernstein. 1994. *Psychometric Theory*. McGraw-Hill.
- OpenAI. 2022. "Introducing ChatGPT." <https://openai.com/blog/chatgpt>.
- Pennefather, P. P. 2023. *Creative Prototyping With Generative AI: Augmenting Creative Workflows With Generative AI*. Ca, Apress.
- Pinski, M., T. Hofmann, and A. Benlian. 2024. "AI Literacy for the Top Management: An Upper Echelons Perspective on Corporate AI Orientation and Implementation Ability." *Electronic Markets* 34: 24.
- Prajogo, D. I., and P. K. Ahmed. 2006. "Relationships Between Innovation Stimulus, Innovation Capacity, and Innovation Performance." *R&D Management* 36: 499–515.
- PTC. 2024. "Generative Design Helps You Deliver Your Best Designs in Less Time." <https://www.ptc.com/en/technologies/cad/generative-design>.
- Rafferty, A. E., N. L. Jimmieson, and A. A. Armenakis. 2013. "Change Readiness: A Multilevel Review." *Journal of Management* 39: 110–135.
- Rajapathirana, R. P. J., and Y. Hui. 2018. "Relationship Between Innovation Capability, Innovation Type, and Firm Performance." *Journal of Innovation & Knowledge* 3: 44–55.
- Ringle, C. M., S. Wende, and J.-M. Becker. 2024. *SmartPLS 4*. SmartPLS.
- Sambasivan, N., S. Kapania, H. Highfill, D. Akrong, P. Paritosh, and L. M. Aroyo. 2021. "Everyone Wants to Do the Model Work, Not the Data Work: Data Cascades in High-Stakes AI." In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery.
- Shahrasbi, N., and G. Paré. 2014. "Rethinking the Concept of Organizational Readiness: What Can IS Researchers Learn From the Change Management Field?" 20th Americas Conference on Information Systems (AMCIS), Savannah.
- Shahrasbi, N., and G. Paré. 2015. "Inside the 'Black Box': Investigating the Link Between Organizational Readiness and IT Implementation Success." ICIS 2015.
- Shelton, R. 2009. "Integrating Product and Service Innovation." *Research-Technology Management* 52: 38–44.
- Sofka, W., and C. Grimpe. 2010. "Specialized Search and Innovation Performance – Evidence Across Europe." *R&D Management* 40: 310–323.
- Steinbauer, G., M. Kandlhofer, T. Chklovski, F. Heintz, and S. Koenig. 2021. "A Differentiated Discussion About AI Education K-12." *KI - Künstliche Intelligenz* 35: 131–137.
- Stratman, J. K., and A. V. Roth. 2002. "Enterprise Resource Planning (ERP) Competence Constructs: Two-Stage Multi-Item Scale Development and Validation." *Decision Sciences* 33: 601–628.
- Sutanonpaiboon, J., and A. M. Pearson. 2006. "E-Commerce Adoption: Perceptions of Managers/Owners of Small- and Medium-Sized Enterprises (SMEs) in Thailand." *Journal of Internet Commerce* 5: 53–82.
- Tashakkori, A., R. B. Johnson, and C. Teddlie. 2020. *Foundations of Mixed Methods Research: Integrating Quantitative and Qualitative Approaches in the Social and Behavioral Sciences*. Thousand Oaks.
- Teece, D. J. 2007. "Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance." *Strategic Management Journal* 28: 1319–1350.
- Teece, D. J., G. Pisano, and A. Shuen. 1997. "Dynamic Capabilities and Strategic Management." *Strategic Management Journal* 18: 509–533.
- Unilever. 2023. "Unilever and Arzeda Use AI to Develop Performance-Boosting Enzymes." <https://www.unilever.ca/news/2023/unilever-and-arzeda-use-ai-to-develop-performanceboosting-enzymes/>.

- Venkatesh, V., M. G. Morris, G. B. Davis, and F. D. Davis. 2003. "User Acceptance of Information Technology: Toward a Unified View." *MIS Quarterly* 27: 425–478.
- Walzl, A., and B. Buchmayr. 2017. "Topologieoptimierung – Entwicklungswerkzeug für die additive Fertigung." *BHM Berg- und Huttenmannische Monatshefte* 162: 110–116.
- Wang, B., P.-L. P. Rau, and T. Yuan. 2023. "Measuring User Competence in Using Artificial Intelligence: Validity and Reliability of Artificial Intelligence Literacy Scale." *Behaviour & Information Technology* 42: 1324–1337.
- Weeks, W. A., J. Roberts, L. B. Chonko, and E. Jones. 2004. "Organizational Readiness for Change, Individual Fear of Change, and Sales Manager Performance: An Empirical Investigation." *Journal of Personal Selling & Sales Management* 24: 7–17.
- Weiner, B. J., H. Amick, and S.-Y. D. Lee. 2008. "Review: Conceptualization and Measurement of Organizational Readiness for Change: A Review of the Literature in Health Services Research and Other Fields." *Medical Care Research and Review* 65: 379–436.
- Weiner, B. J., A. S. Clary, S. L. Klamann, K. Turner, and A. ALISHAHI-Tabriz. 2020. "Organizational Readiness for Change: What we Know, What we Think we Know, and What we Need to Know." In *Implementation Science 3.0*, edited by B. Albers, A. Shlonsky, and R. Mildon. Springer International Publishing.
- Williams, M. D., N. P. Rana, and Y. K. Dwivedi. 2015. "The Unified Theory of Acceptance and Use of Technology (UTAUT): A Literature Review." *Journal of Enterprise Information Management* 28: 443–488.
- Wold, H. 1982. "Soft Modeling: The Basic Design and Some Extensions." In *Systems Under Indirect Observation, Part II*, edited by K. G. Jöreskog and H. Wold. North-Holland.
- Zahra, S. A., and G. George. 2002. "Absorptive Capacity: A Review, Reconceptualization, and Extension." *Academy of Management Review* 27: 185–203.
- Zimmerling, C., C. Poppe, and L. Kärger. 2019. "Virtuelle Produktentwicklung Mittels Simulationsmethoden und KI." *Lightweight Design* 12: 12–19.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** radm70038-sup-0001-supinfo.docx.